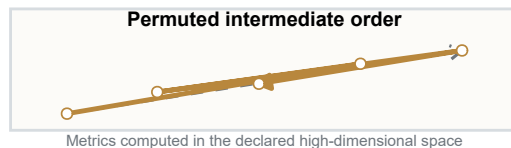
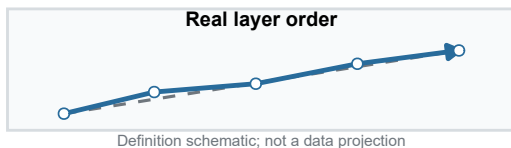
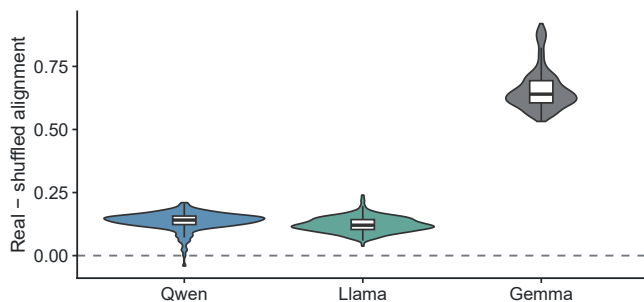


Real trajectories show geometric bias without following exact geodesics

a

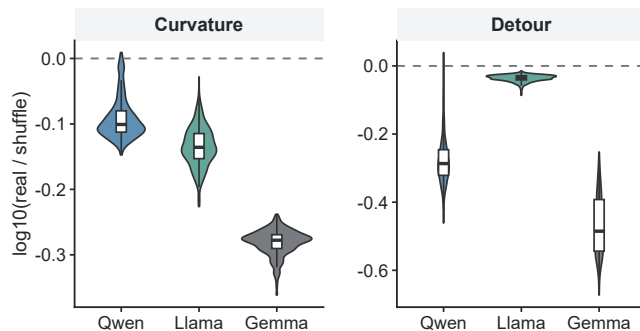


b Alignment gain is heterogeneous but positively shifted



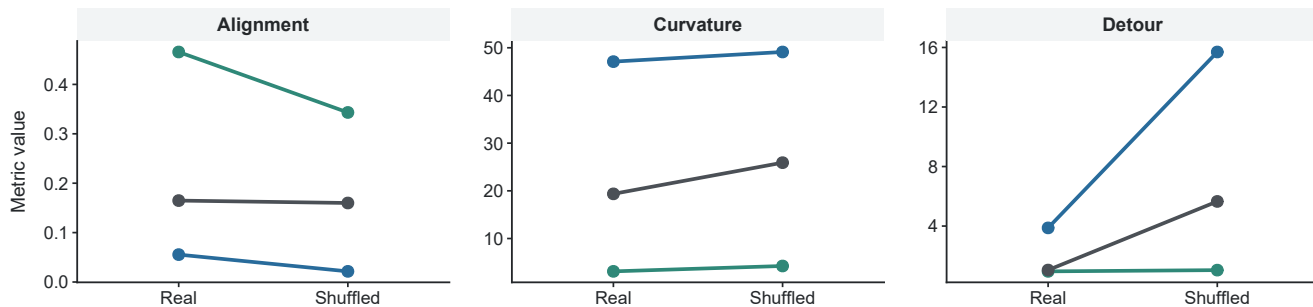
c Real trajectories are less detoured and less curved

Negative values favour real trajectories

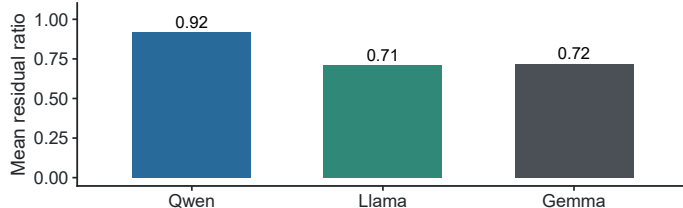


d Selected-window summaries preserve checkpoint differences

Qwen Llama Gemma



e Large residuals reject exact-geodesic motion



f

Revised conclusion

Geodesic-biased flow

Not an exact geodesic or shortest path